IPCON® PPS CF

Carbon Fiber Reinforced Polyphenylene sulfide Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	0.75 kg ± 0.5%
Filament Length	250 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or high-temp PC (max heat resistance: 140 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 110 mm
Recommendation for body protection during use	This material contains carbon fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.26 g/cm³
Shrinkage Rate*	Specially-defined	0.19% - 0.41%
Surface Hardness	Shore hardness	84D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.05%
Melt Index	340 °C, 2.16 kg	20.6 g/10 min
Melting Temperature	DSC, 10 °C/min	284 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	268 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	264 °C
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	200 - 220 °C (Related to load, size, shape, wall thickness, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	93 MPa
Tensile Strength Z	ISO 527, GB/T 1040	46 MPa
Young's Modulus XY	ISO 527, GB/T 1040	8347 MPa
Young's Modulus Z	ISO 527, GB/T 1040	3026 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	1.8%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	0.7%

Flexural Strength XY	ISO 178, GB/T 9341	137 MPa
Flexural Strength Z	ISO 178, GB/T 9341	62 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	7402 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2885 MPa
Impact Strength XY	ISO 179, GB/T 1043	22.3 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	8.8 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu H2S printer (nozzle temperature: 345 °C, bed temperature: 120 °C, printing speed: 100 mm/s) and IPCON PPS CF (Black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. High stiffness, high strength, easy to print (compatible with Bambu H2D, H2S, H2C, and QIDI Q2, Q1 Pro, and other printers that have 350 °C nozzle temperature and positive chamber temperature, as well as AMS or CFS), top-grade flame retardancy (can meet UL94 V-0 @0.8 mm/1.6 mm/3.2 mm), ultra temperature resistance (long-term temperature resistance of 200 °C), superb water resistance and solvent resistance
2. Suitable for automotive engines and fluid systems, aerospace, chemical industry, medical disinfection equipment and other application scenarios that require flame retardancy, heat resistance or water resistance and solvent corrosion resistance.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 140 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 120 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PPS
Compatible Printer Type	Enclosed-frame (60 to 120 °C chamber temperature is recommended)
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.6 mm (recommended) / 0.4 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	320 - 360 °C
Bed Temperature	100 - 120 °C
Chamber Temperature	60 - 120 °C
Fan Speed	0% - 40%
Max Print Speed	300 mm/s

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